

Daheng Yin

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Education

Simon Fraser Univ.

Ph.D. in Computing Science
Advisor: Prof. Jiangchuan Liu
2023.08 -- Present

Southeast Univ.

M.Eng. in Computer Science
Advisor: Prof. Fang Dong
2020.08 -- 2023.07

Jiangnan Univ.

B.Eng. in IoT Engineering
Advisor: Prof. Ya Guo
2016.08 -- 2020.07
GPA: 3.63, Rank 9/141

Research Areas

- Volumetric Video Streaming
- Dynamic 3D Reconstruction
- Embodied AI Systems

Links

Homepage:
yindaheng98.top
Github: [@yindaheng98](https://github.com/yindaheng98)

Research Profile

- I build real-time 4D spatial perception infrastructure for Physical AI: systems that re-construct, compress, and stream photorealistic 4D spatial representations of dynamic physical environments, enabling Physical AI systems to perceive the physical world, interact with it, and access it remotely.

Selected Publications

CAGS: Color-Adaptive Volumetric Video Streaming with Dynamic 3D Gaussian Splatting SIGGRAPH, 2026

- **Daheng Yin**, Yili Jin, Jianxin Shi, Isaac Ding, Miao Zhang, Fangxin Wang, Zhaowu Huang, Cong Zhang, Jiangchuan Liu, Fang Dong
- We propose color-adaptive 3DGS streaming for remote physical environments: compresses dynamic 3DGS volumetric video into vector-quantized LoD representations and restores color fidelity on the client, preserving photorealistic access under fluctuating bandwidth.

TrackerSplat: Exploiting Point Tracking for Fast and Robust Dynamic 3D Gaussians Reconstruction SIGGRAPH Asia, 2025

- **Daheng Yin**, Isaac Ding, Yili Jin, Jianxin Shi, Jiangchuan Liu
- High-performance dynamic 3D scene reconstruction for fast-moving scenes with point tracking: lifts cross-frame point trajectories onto 3D Gaussians and guides their translation, rotation, and scale compensation, reducing drift, artifacts, and temporal flickering.

FSVFG: Towards Immersive Full-Scene Volumetric Video Streaming with Adaptive Feature Grid ACM Multimedia, 2024

- **Daheng Yin**, Jianxin Shi, Miao Zhang, Zhaowu Huang, Jiangchuan Liu, Fang Dong
- Adaptive feature-grid streaming for large-scale full-scene volumetric video: represents wide-area dynamic scenes with compact feature grids and selectively transmits features and residuals, balancing scene coverage, visual quality, and transmission cost.

Contests & Honors

2025	SIGGRAPH Asia Volumetric Video Workshop	Compression Track 3rd
2022	TensorRT Hackathon (NVIDIA & TIANCHI)	Winner Prize
2021	TensorRT Hackathon (NVIDIA & TIANCHI)	Rank 4/48
2020	Outstanding Graduate, Jiangnan University	
2018	National College Mathematical Contest in Modeling	2nd Prize (National)
2017	China National Scholarship (2016-2017)	
2017	9th National College Mathematical Contest	2nd Prize (Provincial)
2017	14th Jiangsu College Mathematical Contest	1st Prize